Sensitivity-Gradient Tipping, MPC-Guided Policy Search and Suspension Integrity for Cable-Only Locomotion of a Six-Bar Tensegrity Rover

[TODO: First Author][1] and [TODO: Second Author][2]

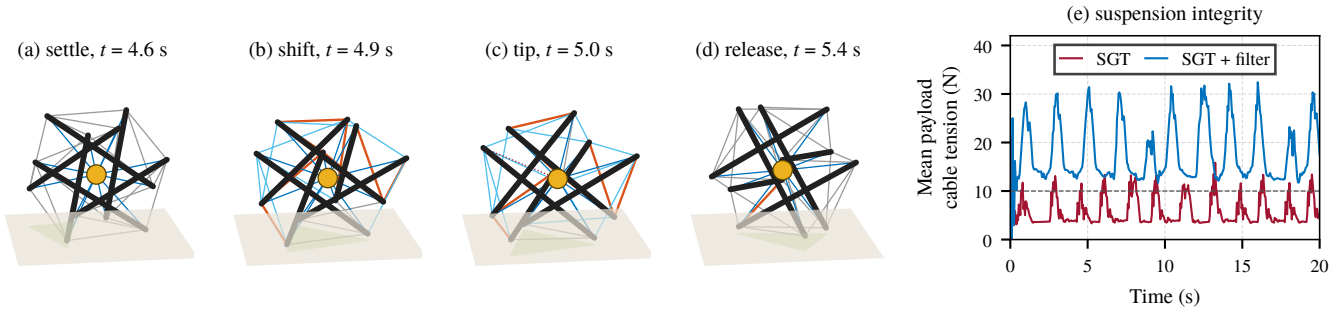


Fig. 1. One cable-only tip of the six-bar rover under Sensitivity-Gradient Tipping with the suspension-integrity filter, rendered from the simulator: (a) settled on a support face (green), (b) the cables selected by the sensitivity contract (orange) or lengthen (light blue) and shift the payload (yellow), (c) the centre of mass crosses the support edge, (d) cables released on the new face. Payload cables are blue when taut and dotted red when slack. (e) Mean payload-cable tension over 20 s: without the filter the suspension stays below 10 N (dashed) and is partly slack; with it the payload remains suspended.

***Abstract*—Six-bar tensegrity rovers roll by deforming their cable network, but their gaits are developed in simulators that can inject energy, and their suspended payloads are rarely examined. We present three cable-only controllers on an energy-audited simulator. Sensitivity-Gradient Tipping (SGT) differentiates the pinned quasi-static equilibrium and reaches 0.091 m/s, where support-face LQR, iLQR and MPC schedulers advance at most 0.10 m in 20 s. MPC + GPS, an MPC-guided policy search whose teacher reduces exactly to SGT, learns a cable policy reaching 89 % of SGT's progress on held-out conditions. A suspension-integrity filter with discrete-time barrier conditions reduces payload-cable slack from 35.1 % to 2.7 % of the time and increases SGT's progress by 48 %. Code: https://github.com/kapildev1012/3d.**

I. Introduction

Six-bar spherical tensegrities combine low mass with high impact tolerance and have been studied as planetary landers and rovers [1], [2]. They move by *punctuated rolling*: changing cable rest lengths deforms the shell until the centre of mass (COM) leaves the triangular support face and the structure tips onto a neighbouring face (Fig. 1). Because hardware is expensive and fragile, gaits are developed in simulation [3], [4] and transferred later, which makes two questions central: does the simulated motion come from the cables, and does the gait leave the payload it carries intact?

General engines such as MuJoCo and Bullet [5], [6] and dedicated tensegrity simulators [3] treat bars as rigid bodies or stiff penalty springs whose drift is removed by stabilization [7] or by position projection (SHAKE/RATTLE [8], [9]); contacts are often clamped, and interactive simulators add "assist" torques so that a gait looks convincing. Each of these changes the mechanical energy without a physical work term, and structure-preserving integrators [10] do not by themselves expose such sources. On the control side, rolling gaits have been obtained from hand-designed cable sequences, kinodynamic planning over gait primitives [2] and reinforcement learning, including guided policy search on SUPERball [4]. Guided policy search trains a policy by supervised learning on local trajectory optimizers that are kept close to it [11], [12]; its MPC-guided variant uses a model-predictive teacher [13], and PLATO and DAgger train on the learner's own state distribution [14], [15]. Control barrier functions enforce safety by a minimal correction of a nominal input [16], [17], but we are not aware of their use for the suspension of a tensegrity payload. Classic support-face gaits, finally, contract a fixed cable set; with a realistic stroke and a heavy payload such a primitive may not move the COM far enough to tip, however its effort is scheduled.

Building on an *energy-audited* simulator—a non-minimal Lagrangian model [18] with exact bar multipliers, passive contact and a work ledger that measures, before correction, the energy that each step creates without a booked cause—we make three contributions (Fig. 2): (1) *Sensitivity-Gradient Tipping* (SGT), a cable-only gait that differentiates the pinned quasi-static equilibrium of shell and payload, including the geometric stiffness of the compressed bars, and drives the COM across a support edge; (2) *MPC + GPS*, an MPC-guided policy

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[1][TODO: Department, Institution, City, Country.]
[2][TODO: Department, Institution, City, Country.]
Corresponding author: [TODO: email].

search—throughout, GPS denotes Guided Policy Search—whose receding-horizon QP teacher on the SGT sensitivity reduces exactly to SGT in a limit and trains a permutation-equivariant per-cable policy; and (3) the observation that the payload suspension of a rolling six-bar rover loses its pretension, with a *suspension-integrity filter* that minimally corrects any tipping controller using the same sensitivities. All methods are evaluated on a seed-locked protocol (114 runs for SGT and five baselines, 24 for policy training and evaluation, 40 for the filter), and every number in the text is generated from the run files. All claims are restricted to simulation, and the negative results for the baselines concern the support-face primitive they schedule, not the optimal-control methods as such.

II. Energy-Audited Shell Dynamics

A. Coordinates, Mass and Constraints

The shell has $n = 12$ nodes $\boldsymbol{q} \in \mathbb{R}^{36}$, six bars connecting node pairs (i, j) and 24 outer cables; the payload core is a point mass m_c suspended by 12 cables. Bar k of length l_k, mass m_k and radius r_k is a uniform cylinder, which gives the consistent block mass matrix (times I_3)

$$\boldsymbol{M}_k = \begin{bmatrix} a_k + m_i & b_k \\ b_k & a_k + m_j \end{bmatrix},\ a_k = \tfrac{m_k}{4} + \tfrac{J_k}{l_k^2},\ b_k = \tfrac{m_k}{4} - \tfrac{J_k}{l_k^2}, \tag{1}$$

with $J_k = \frac{m_k}{12}(3r_k^2 + l_k^2)$ and node masses m_i. The bars are node-disjoint, so $\boldsymbol{M}$ is block diagonal with closed-form inverse. The bar constraints are $R_k(\boldsymbol{q}) = \|\boldsymbol{b}_k\|^2 - l_k^2 = 0$, $\boldsymbol{b}_k = \boldsymbol{q}_j - \boldsymbol{q}_i$, with Jacobian $\boldsymbol{R}_q$.

B. Equations of Motion and Correction

With $\boldsymbol{F}$ collecting gravity, cable, damping, contact and payload forces, the index-1 form of $\boldsymbol{M}\ddot{\boldsymbol{q}} = \boldsymbol{F} + \boldsymbol{R}_q^\top \boldsymbol{\lambda}$ gives the multipliers in closed form,

$$\boldsymbol{\lambda} = -\big(\boldsymbol{R}_q \boldsymbol{M}^{-1} \boldsymbol{R}_q^\top\big)^{-1}\big(\boldsymbol{\xi}_2 + \boldsymbol{R}_q \boldsymbol{M}^{-1} \boldsymbol{F}\big), \quad \xi_{2,k} = 2\|\dot{\boldsymbol{b}}_k\|^2, \tag{2}$$

and the bar compression $P_k = 2\lambda_k l_k$ ($\Delta t = 2\,\mathrm{ms}$, semi-implicit Euler). After every step the minimum-norm corrector of [18] iterates, for at most ten iterations,

$$\boldsymbol{r} = \big[\boldsymbol{R};\ \dot{\boldsymbol{R}};\ E + V_c - (E^\star + \delta W)\big], \tag{3a}$$

$$\boldsymbol{d} = -\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{A}^\top)^{-1}\boldsymbol{r}, \quad \boldsymbol{A} = \partial \boldsymbol{r}/\partial[\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W], \tag{3b}$$

$$[\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W] \leftarrow [\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W] + \boldsymbol{d}, \tag{3c}$$

where $E^\star$ is the energy predicted by the work ledger and the correction work δW is booked explicitly instead of being hidden. Our extension is the contact potential V_c in the energy function. The energy row is enforced only while $|E - E^\star| \le \max(0.5\,\mathrm{J}, 0.02|E|)$, so a gross modelling error is reported rather than corrected.

C. Passive Contact

For penetration $p = \max(0, h(x, y) - z)$ into terrain h, the normal force is the Hunt–Crossley law [19] $\boldsymbol{f}_n = k\,p^{3/2}(1 + \alpha\dot{p})\hat{\boldsymbol{n}}$, whose elastic part has potential $V_c = \frac{2}{5}k\,p^{5/2}$ and whose damping only dissipates; friction is a regularized Coulomb law $\boldsymbol{f}_t = -\mu\|\boldsymbol{f}_n\|\boldsymbol{v}_t/\sqrt{\|\boldsymbol{v}_t\|^2 + \epsilon^2}$. No clamps, impulses or velocity projections are used.

D. Work Ledger

The ledger stores $E = T + V_g + V_c$ and the trapezoidal work W_c of every force category; projections are booked as the energy change they cause. The certificate and the per-step drift are

$$\varepsilon(t) = E(t) - E_0 - \textstyle\sum_c W_c(t), \qquad \delta_k = \varepsilon_k^- - \varepsilon_{k-1}, \tag{4}$$

where ε_k^- is the residual before the energy row is applied: δ_k is the energy that step k created or destroyed without a booked cause. For a correct model it is discretization error and shrinks with Δt; an unbooked force would not. We also report the numerical work $|W_{\mathrm{num}}|$ of projections and corrector and the *aid share* of the energy input supplied by non-physical categories. A run is *admissible* if the aid share is zero and $\max_k |\delta_k|$ and $|W_{\mathrm{num}}|$ are small compared with the actuator work.

III. Sensitivity-Gradient Tipping

A. Quasi-Static Sensitivity

Between tips the rover rests on a support face $\mathcal{F}$ of three sticking nodes. Augmenting $\boldsymbol{q}$ with the core ($n' = 13$ nodes) and linearizing the static equilibrium with respect to the rest-length offsets $\boldsymbol{u} \in \mathbb{R}^{36}$ ($u > 0$ shortens) gives

$$\begin{bmatrix} \boldsymbol{K} & \boldsymbol{A}^\top \\ \boldsymbol{A} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \delta\boldsymbol{q} \\ \delta\boldsymbol{\mu} \end{bmatrix} = \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{0} \end{bmatrix} \delta\boldsymbol{u}, \quad \boldsymbol{A} = \begin{bmatrix} \boldsymbol{R}_q \\ \boldsymbol{E}_{\mathcal{F}} \end{bmatrix}, \tag{5}$$

where $\boldsymbol{E}_{\mathcal{F}}$ pins the support nodes and $\boldsymbol{B}$ contains $\pm k\boldsymbol{e}$ for every taut cable. $\boldsymbol{K}$ assembles $k\boldsymbol{e}\boldsymbol{e}^\top + \frac{t}{L}(\boldsymbol{I} - \boldsymbol{e}\boldsymbol{e}^\top)$ for a taut cable and the *negative* geometric stiffness $-\frac{P}{L}(\boldsymbol{I} - \boldsymbol{e}\boldsymbol{e}^\top)$ for a bar under compression P; axial bar rigidity is a constraint, and a small Tikhonov term keeps the system regular. One LU factorization gives the displacement sensitivities $\boldsymbol{X} = \partial\boldsymbol{q}/\partial\boldsymbol{u}$ and the COM sensitivity $\boldsymbol{S} = \frac{1}{m_{\mathrm{tot}}}\sum_i m_i \boldsymbol{X}_i$.

B. Edge Selection and Command

For an edge e of $\mathcal{F}$ with outward normal $\boldsymbol{n}_e$ and midpoint $\boldsymbol{p}_e$, the tipping margin is $m_e = -(\boldsymbol{c} - \boldsymbol{p}_e)^\top \boldsymbol{n}_e$ and the gradient of the COM shift across it is $\boldsymbol{g}_e = \boldsymbol{S}_{xy}^\top \boldsymbol{n}_e$. Under box limits $u_k \in [\underline{u}_k, \bar{u}_k]$ the achievable shift is $s_e = \sum_k \max(g_{e,k}\bar{u}_k, g_{e,k}\underline{u}_k)$. Among edges aligned with the travel direction ($\boldsymbol{n}_e^\top \boldsymbol{d} \ge 0.25$), SGT locks for the cycle the edge with the largest reachable margin $s_e - m_e$ and commands

$$u_k = \bar{u}_k\,\mathrm{clip}\big(g_{e,k}/\|\boldsymbol{g}_e\|_\infty, -\rho, 1\big), \tag{6}$$

normalized separately for shell and payload cables. A state machine (Settle→Shift→Tip→Release) re-plans at 20 Hz until the support face changes. A support-face primitive contracts a fixed set whose effect is mostly absorbed by elastic

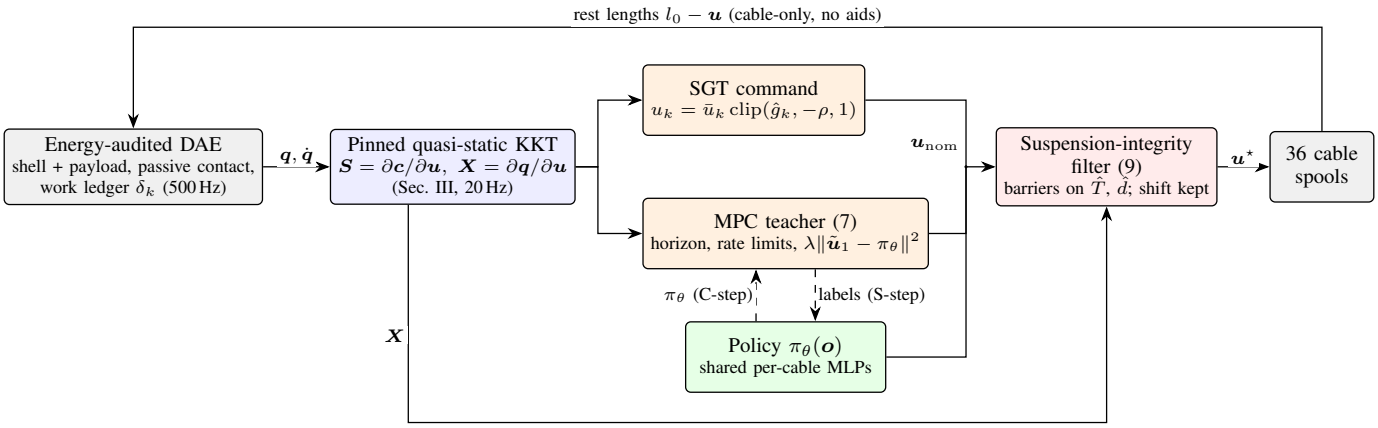


Fig. 2. Architecture. The energy-audited simulator (Sec. II) provides the state; one factorization of the pinned quasi-static KKT system gives the sensitivities used by all three components: SGT (Sec. III), the MPC teacher of MPC + GPS and its learned policy (Sec. IV; dashed: training), and the suspension-integrity filter applied to any nominal command (Sec. V). The rover is actuated only through cable rest lengths.

deformation; the constrained sensitivity instead tells which cables move the COM, and the compressed bars soften the structure in exactly the bending directions that tipping needs.

IV. MPC + GPS: MPC-Guided Policy Search

A. MPC Teacher

With $\hat{\boldsymbol{g}}$ the gradient of (6) normalized per group and $\tilde{\boldsymbol{u}} = \boldsymbol{u}/\bar{\boldsymbol{u}}$, the teacher solves, over N steps of Δt,

$$\begin{aligned} \min_{\tilde{\boldsymbol{u}}_{1:N}} \ & \textstyle\sum_{k=1}^{N} \big(-\hat{\boldsymbol{g}}^\top \tilde{\boldsymbol{u}}_k + \tfrac{r}{2}\|\tilde{\boldsymbol{u}}_k\|^2 + \tfrac{r_\Delta}{2}\|\tilde{\boldsymbol{u}}_k - \tilde{\boldsymbol{u}}_{k-1}\|^2 \big) \\ & + \tfrac{\lambda}{2}\|\tilde{\boldsymbol{u}}_1 - \pi_\theta(\boldsymbol{o})\|^2 \\ \text{s.t. } & \underline{\tilde{u}} \le \tilde{\boldsymbol{u}}_k \le 1, \quad |\tilde{\boldsymbol{u}}_k - \tilde{\boldsymbol{u}}_{k-1}| \le \rho_s, \end{aligned} \tag{7}$$

with $\tilde{\boldsymbol{u}}_0$ the current spool state and ρ_s the spool speed times Δt ($N = 6$, $\Delta t = 50$ ms, $r = 1$, $r_\Delta = 0.2$). The problem is separable per actuator and is solved by projected gradient at 20 Hz.

Remark 1. For $N = 1$, $r = 1$, $r_\Delta = \lambda = 0$ and an inactive rate bound, each actuator minimizes $-\hat{g}_k\tilde{u}_k + \frac{1}{2}\tilde{u}_k^2$ on $[\underline{\tilde{u}}_k, 1]$, whose solution $\operatorname{clip}(\hat{g}_k, \underline{\tilde{u}}_k, 1)$ is the SGT command (6). The teacher thus generalizes SGT with a horizon, spool-rate limits and smoothing, and $\lambda > 0$ adds the policy-proximity term of guided policy search—the quadratic form of the KL constraint between fixed-covariance Gaussian policies.

B. Policy Observations

The policy π_θ does not use the KKT model. Each outer cable is described by 20 local features in the frame of the travel direction: midpoint relative to the COM, end-node clearances, signed distance to the locked edge and along it, membership of its end nodes in face and edge, the sign-invariant orientation terms $(\boldsymbol{e}^\top \boldsymbol{n}_e)e_z$, $(\boldsymbol{e}^\top \boldsymbol{n}_e)^2$ and e_z^2, current offset and strain, and global context (COM velocity, gait state, margin, time in state); the payload cables have 19 analogous features. One MLP (2×48 units) is shared by all outer cables and one (2×36) by all payload cables, so the policy is equivariant to the cable numbering and independent of the face the rover rests on.

C. Training

Starting from $\mathcal{D}_0$ collected with the unregularized teacher, iteration i alternates

$$\text{C-step: roll out (7) with } \lambda_i, \text{ and } \pi_{\theta_i} \text{ itself,} \tag{8a}$$

$$\mathcal{D}_{i+1} = \mathcal{D}_i \cup \{(\boldsymbol{o}, \tilde{\boldsymbol{u}}^\star_{\lambda=0}(\boldsymbol{o}))\}, \tag{8b}$$

$$\text{S-step: } \theta_{i+1} = \arg\min_\theta \textstyle\sum_{\mathcal{D}_{i+1}} \|\pi_\theta(\boldsymbol{o}) - \tilde{\boldsymbol{u}}^\star\|^2, \tag{8c}$$

i.e. every visited state is labelled with the unregularized teacher as in PLATO [14], while the executed actions keep the data close to the learner's own state distribution. We use $\lambda_i \in \{0.5, 1, 2, \dots\}$, four 15 s episodes per iteration (two of them executing the policy) over Levels 1 and 3 with straight and $\pm$8–10 m offset goals, and 8 iterations (114k labelled actuator samples, Adam).

D. Deployment

At run time the policy replaces the per-tick KKT solve while the state machine is kept. Choosing the edge by alignment alone sent the policy to edges it could not tip over, so the deployed controller solves (5) once per tip cycle to pick the edge with the largest reachable margin, as the teacher does, and excludes edges that aborted from the same face.

V. Suspension-Integrity Filter

A. Observation

Gravity and the outer-cable actuation compress the shell while it rolls: under SGT the mean node height drops from 0.467 to 0.394 m (16 %). The node-to-payload distances shrink with it, and the payload cables, pretensioned to 50 N, carry 5.7 N on average; at least one of the 12 is slack in 100 % of the samples (Fig. 1e). The "suspended" payload is loose in the cage for most of the motion.

B. Filter

Given the command $\boldsymbol{u}_{\mathrm{nom}}$ of any tipping controller, the filter returns

$$\begin{aligned}\boldsymbol{u}^{\star} = \arg\min_{\boldsymbol{u}}\ & \tfrac{1}{2}(\boldsymbol{u}-\boldsymbol{u}_{\mathrm{nom}})^{\top}\boldsymbol{W}(\boldsymbol{u}-\boldsymbol{u}_{\mathrm{nom}}) \\ \text{s.t. } & \hat{h}_c(\boldsymbol{u}) \geq (1-\gamma)\, h_c, \quad c \in \mathcal{C}, \\ & \hat{\boldsymbol{g}}^{\top}(\boldsymbol{u}-\boldsymbol{u}_0) \geq \hat{\boldsymbol{g}}^{\top}(\boldsymbol{u}_{\mathrm{nom}}-\boldsymbol{u}_0), \quad \underline{\boldsymbol{u}} \leq \boldsymbol{u} \leq \bar{\boldsymbol{u}},\end{aligned} \tag{9}$$

where $\boldsymbol{u}_0$ is the current command and $\hat{h}_c$ is the first-order prediction of a safety function from $\boldsymbol{X}$. For a cable between nodes i, j with stiffness k, rest length l_0 and direction $\boldsymbol{e}$, $\hat{T}(\boldsymbol{u}) = k\big(L - l_0 + u_{\mathrm{self}} + \boldsymbol{e}^{\top}(\boldsymbol{X}_j - \boldsymbol{X}_i)(\boldsymbol{u}-\boldsymbol{u}_0)\big)$. $\mathcal{C}$ contains payload-cable tension $T \geq 10\,\mathrm{N}$, outer-cable tension $T \leq 150\,\mathrm{N}$ and payload-to-rod clearance $d \geq 0.08\,\mathrm{m}$ (core radius 0.05 m), the latter linearized at the closest point of each rod. The barrier condition is the discrete-time form of [17]: a satisfied constraint stays satisfied and a violated one recovers at rate $\gamma = 0.3$ per step. The second constraint preserves the predicted COM shift: without it the filter cancels the payload-cable slackening that SGT uses to tip; with it the tipping effort is re-routed, mainly to the leading payload cables. It is dropped if it conflicts with the barriers. $\boldsymbol{W}$ weights outer cables ten times more than payload cables, and the QP is solved by accelerated projected gradient on the dual [20] with the box exact in the primal.

Remark 2. The conditions hold for the quasi-static prediction $\hat{h}$. Tumbling and impacts are inertial events that this prediction does not contain, so the closed-loop results of Section VI-E are empirical, and we report where they fail.

VI. Results

A. Experimental Setup

Platform. Shell radius 0.5 m (bar length 0.851 m), 12 nodes of 0.2 kg, six bars of 0.3 kg, payload 1.6 kg (5.8 kg in total), cable stiffness 1200 N/m (payload cables 1600 N/m). Every method uses the same actuator box: stroke 0.15 m (30 % of the 0.49 m rest length) and 0.20 m/s spool speed. Physics runs at 500 Hz, controllers at 20 Hz, on the DAE core with (3), passive contact and zero aids, on terrain Level 1 for 20 s unless stated otherwise.

Baselines. Support-face heuristic: the fixed contraction primitive. *LQR, iLQR, QP-MPC*: the controllers of the original code base, which solve a two-state reduced-order rolling model for a normalized effort that scales the same primitive. *Legacy+assist*†: the original penalty+SHAKE engine with roll couple, traction, grip and corridor aids, reported as an inadmissible reference.

Scenarios and metrics. Lateral impulse 0.5 m/s at 6 s; push 10 N for 2 s; Gaussian, Laplace and outlier (2 %, 10×) sensor noise; cable stiffness $-20\,\%$, friction $-40\,\%$, payload $+30\,\%$; actuation delay 100 ms. We report mean speed $\bar{v}$, cross-track error $e(t) = x_c(t) - x_c(0)$ (RMS and max), success ($\Delta y \geq 1\,\mathrm{m}$ in 20 s), effort $\int \boldsymbol{u}^{\top}\boldsymbol{u}\,dt$, cost of transport $\mathrm{CoT} = W_{\mathrm{act}}/(mg\Delta y)$, controller time and the ledger quantities of Section II; statistics are mean ± std over seeds 1–5 (nominal) or 1–3. The policy is evaluated on 4 held-out conditions (unseen seeds, Levels 1 and 3, straight and ±6 m goals); the filter on 10 paired conditions (three seeds on Level 1, two on Level 3, push, impulse, payload $+30\,\%$, delay, stiffness $-20\,\%$) with payload-cable slack time and tension, payload rattle (RMS position relative to the shell centroid), 99th-percentile payload acceleration, minimal payload–rod clearance and maximal outer-cable tension, measured after the first second.

B. Physical Consistency

Table II compares the cores under the same SGT command sequence. Three observations stand out. *(i) Rigidity.* The DAE core keeps the bar lengths at 2.3×10^{-11} m; the penalty+SHAKE core reaches 1.3×10^{-7} m and, with the interactive aids, 1.9×10^{-4} m. *(ii) Hidden energy exchange.* To keep the bars rigid, the penalty+SHAKE core removes 6.8×10^{3} J by position projection while its cables do a comparable amount of work, i.e. the cables mostly pump energy into the penalty bars that the projection then discards. The DAE core changes only 7.3×10^{-3} J numerically. With the original locomotion aids, 50 % of the energy input comes from non-physical forces, which explains its high speed (0.522 m/s) in Table I. *(iii) Drift.* Before correction, both cores show a per-step energy mismatch that is first-order discretization error: for seed 1, halving Δt from 2 to 1 ms reduces the accumulated $\sum_k |\delta_k|$ from 15.4 to 7.6 J for the DAE core and from 11.0 to 5.5 J for penalty+SHAKE, and the largest single-step mismatch of the DAE core (at the initial landing) from 7.6×10^{-2} to 1.9×10^{-2} J. The energy row removes this drift every step, so ε stays at the solver tolerance (1.0×10^{-9} J). Without the energy row the residual accumulates to 1.5×10^{1} J. We stress that the energy row corrects an integration error of known order; it does not by itself prove that the model is complete—that is the role of δ_k, which would expose an unbooked force as a non-convergent drift. On rugged Level 7 the corrector occasionally reaches its 10-iteration cap during contact transitions, leaving a transient residual of up to 4×10^{-2} J (0.2 % of E) that is removed in the following steps (Table VII).

C. Cable-Only Locomotion with SGT

1) Nominal: with identical actuator limits and all aids disabled, SGT advances 1.81 m in 20 s (0.091 ± 0.026 m/s, 100 % success; Table I, Fig. 3), about 25 times the speed of the support-face heuristic. The LQR, iLQR and QP-MPC schedulers optimize a scalar effort over the same support-face primitive and therefore inherit its limitation: none of them leaves the launch pad (at most 0.10 m), and their cost of transport is an order of magnitude higher. This is a statement about the primitive, which cannot shift the COM of the 5.8 kg system over an edge within a 0.15 m stroke, rather than about the optimizers. SGT plans in 0.68 ms on average (worst 17.4 ms) in single-threaded JavaScript, well inside its 50 ms control period. Its main weakness is heading: the greedy per-cycle edge choice produces a zig-zag track with a cross-track

TABLE I

NOMINAL CABLE-ONLY LOCOMOTION ON LEVEL 1 (20 S, 5 SEEDS, MEAN ± STD). ALL METHODS USE THE LAGRANGIAN DAE CORE, PASSIVE CONTACT AND IDENTICAL ACTUATOR LIMITS.

Method	$\bar{v}$ (m/s)↑	RMSE (m)↓	max $\lvert e\rvert$ (m)↓	$\int u^\top u\,dt$↓	CoT↓	t_{solve} (ms)↓	Succ. (%)↑	Aid (%)↓
SGT (proposed)	$\mathbf{0.091 \pm 0.026}$	0.253 ± 0.029	0.473 ± 0.000	$\mathbf{0.334 \pm 0.170}$	$\mathbf{11.4 \pm 6.6}$	0.68 ± 0.10	**100**	0
Heuristic	0.004	0.083	0.160	2.059	142.4	$\mathbf{0.47 \pm 0.25}$	0	0
LQR	0.002	0.129	0.163	2.529	–	0.65 ± 0.06	0	0
iLQR	0.005	0.081	0.168	1.757 ± 0.000	111.8	0.90 ± 0.41	0	0
QP-MPC	0.004	0.084	0.175	1.859 ± 0.000	132.0	0.82 ± 0.05	0	0
Legacy+assist†	0.522 ± 0.023	0.006 ± 0.002	0.017 ± 0.006	0.636 ± 0.060	1.1 ± 0.1	0.15 ± 0.02	100	50

†Legacy engine with external roll couple, traction, grip and corridor aids; not physically admissible and excluded from the bold "best" marks. Success: $\Delta y \geq 1$ m in 20 s. Aid: share of the energy input supplied by non-physical aids. Cross-track errors are not marked because they are only comparable between methods that travel a similar distance.

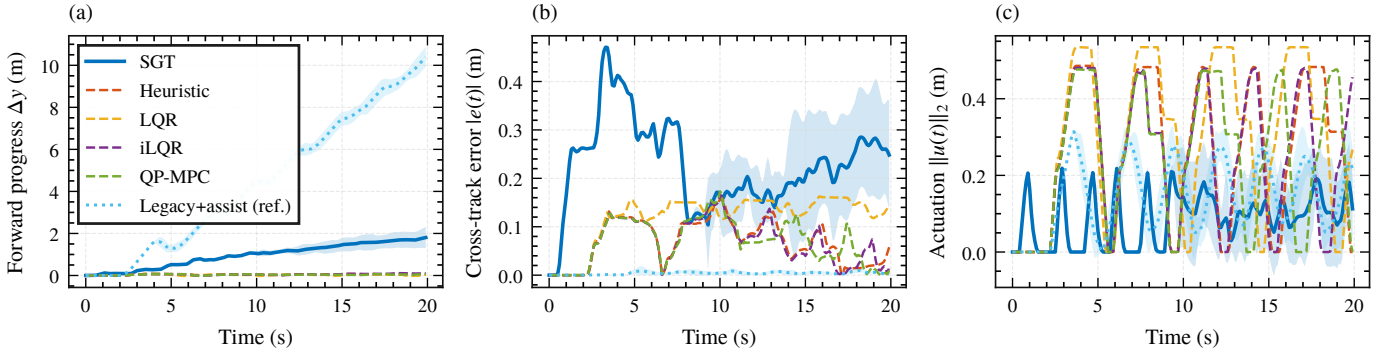


Fig. 3. Nominal cable-only locomotion (5 seeds, mean $\pm 1\sigma$). (a) Forward progress. (b) Cross-track error. (c) Actuation norm. Solid: proposed; dashed: baselines; dotted: inadmissible reference.

TABLE II

PHYSICAL CONSISTENCY OF THE SIMULATOR UNDER THE SGT GAIT (LEVEL 1, 20 S). $\varepsilon = E - E_0 - \sum W$ IS THE ENERGY NO BOOKED WORK EXPLAINS *after* CORRECTION (ENFORCED BY THE ENERGY ROW); δ_k IS THE SAME QUANTITY FOR STEP k *before* CORRECTION, I.E. THE PER-STEP BOUND ON ANY UNBOOKED ENERGY SOURCE; $|W_{\text{num}}|$ IS THE ENERGY CHANGED BY CONSTRAINT PROJECTIONS AND BY THE CORRECTOR (BOOKED, BUT NUMERICAL). SGT ON LEVEL 1, 20 S; †THE ORIGINAL ENGINE WITH ITS LOCOMOTION AIDS AND SUPPORT-FACE GAIT, FOR REFERENCE.

Method	max $\lvert\varepsilon\rvert$ (J)	max $\lvert\delta_k\rvert$ (J)↓	$\sum \lvert\delta_k\rvert$ (J)	bar err. (m)↓	$\lvert W_{\text{num}}\rvert$ (J)↓	$\bar{v}$ (m/s)	Aid (%)
DAE + Alg. 1, energy row	1.0×10^{-9}	7.6×10^{-2}	1.7×10^{1}	2.3×10^{-11}	7.3×10^{-3}	0.091 ± 0.026	0
DAE + Alg. 1, constr. only	1.5×10^{1}	7.1×10^{0}	3.3×10^{1}	5.9×10^{-11}	3.9×10^{1}	0.104 ± 0.029	0
Penalty + SHAKE (legacy)	1.1×10^{1}	1.1×10^{-3}	1.1×10^{1}	1.3×10^{-7}	6.8×10^{3}	0.026	0
Legacy + assists†	1.1×10^{1}	1.2×10^{-3}	1.1×10^{1}	1.9×10^{-4}	1.5×10^{4}	0.522 ± 0.023	50

RMSE of 0.253 m (full ground tracks are in the released run files).

2) Robustness: table III summarizes the stress tests (3 seeds each). SGT keeps rolling after a 0.5 m/s lateral impulse (0.094 m/s) and under a 10 N, 2 s push (0.085 m/s), and is insensitive to $-40\,\%$ friction (0.104 m/s) and $+30\,\%$ payload mass (0.083 m/s). Softer cables ($-20\,\%$) slow it to 0.057 m/s because the same stroke produces smaller COM shifts, and a 100 ms actuation delay to 0.050 m/s (67 % success). Measurement noise is the main weakness: Gaussian noise reduces the speed to 0.065 m/s, while heavy-tailed noise is damaging—Laplace noise to 0.026 m/s (0 % success) and 2 % outliers to 0.034 m/s. SGT uses the measured node positions both to detect the support face and to assemble $\boldsymbol{K}$, so a single corrupted node can trigger a wrong face or edge; a robust state estimator in front of the planner is required before hardware use.

3) Ablation: table IV isolates the ingredients of SGT (3 seeds). Removing the bars' geometric stiffness is the most harmful change: speed drops from 0.080 to 0.056 m/s, success to 67 % and the cost of transport nearly doubles (23.5 vs. 13.4), confirming that the softening of the compressed bars is what the sensitivity has to capture. The other design choices did not help in our tests: without payload actuation the speed is essentially unchanged (0.076 m/s) at a lower cost of transport (7.7); restricting the command to the 8 most sensitive cables keeps the speed (0.085 m/s) and gives the lowest cost of transport (5.6); and choosing the edge by alignment with the travel direction instead of by reachable margin was not worse (0.102 ± 0.034 m/s). With three seeds these differences are within one standard deviation, so we report the margin-based rule as specified before the experiments and leave the sparse, payload-free variant as the more efficient configuration. Run on the penalty+SHAKE core without aids, the same controller

TABLE III
ROBUSTNESS: DISTURBANCES, SENSOR NOISE, PLANT MISMATCH AND ACTUATOR DELAY (LEVEL 1, 20 S, 3 SEEDS).

Scenario	SGT $\bar{v}$ (m/s)	Heur. $\bar{v}$ (m/s)	SGT RMSE (m)	t_{rec} (s)	SGT succ. (%)
Nominal	0.080 ± 0.029	0.004	0.258 ± 0.040	–	100
Lateral impulse 0.5 m/s	0.094 ± 0.017	0.004	0.266 ± 0.011	0.09	100
Step push 10 N, 2 s	0.085 ± 0.006	-0.001	0.308 ± 0.035	2.00	100
Gaussian sensor noise	0.065 ± 0.016	0.001 ± 0.002	0.219 ± 0.030	–	67
Laplace sensor noise	0.026 ± 0.016	-0.001 ± 0.004	0.158 ± 0.036	–	0
Sensor outliers (2 %, 10x)	0.034 ± 0.023	0.002 ± 0.002	0.167 ± 0.052	–	33
Cable stiffness -20 %	0.057 ± 0.002	0.002	0.226 ± 0.004	–	100
Friction -40 %	0.104 ± 0.024	0.002	0.224 ± 0.051	–	100
Payload mass +30 %	0.083 ± 0.016	0.002	0.249 ± 0.022	–	100
Actuator delay 100 ms	0.050 ± 0.003	0.004	0.192 ± 0.002	–	67

TABLE IV
ABLATION OF THE SGT GAIT AND OF THE PHYSICS CORE (LEVEL 1, 20 S, 3 SEEDS).

Variant	$\bar{v}$ (m/s)↑	tips↑	RMSE (m)↓	CoT↓	Succ. (%)
SGT (full)	0.080 ± 0.029	8.7 ± 3.5	0.258 ± 0.040	13.4 ± 8.4	**100**
w/o payload actuation	0.076 ± 0.005	8.7 ± 0.6	0.242 ± 0.006	7.7 ± 0.0	**100**
w/o feasibility edge choice	0.102 ± 0.034	10.3 ± 0.6	0.260 ± 0.041	8.9 ± 3.8	**100**
w/o bar geometric stiffness	0.056 ± 0.010	5.7 ± 1.2	0.194 ± 0.001	23.5 ± 4.3	67
sparse: 8 cables	0.085 ± 0.002	9.3 ± 0.6	0.286 ± 0.025	$\mathbf{5.6 \pm 0.6}$	**100**
DAE w/o energy row	$\mathbf{0.104 \pm 0.029}$	10.3 ± 2.1	0.242 ± 0.026	9.4 ± 5.1	**100**
legacy SHAKE core	0.026	**14.0**	**0.043**	49.1	0

reaches only 0.026 m/s: the projection that absorbs the cable work also absorbs the deformation that SGT relies on. Results on rougher terrain are in the Appendix (Table VII).

D. MPC + GPS

On the held-out conditions the learned policy advances 1.89 m in 20 s, 89 % of SGT (2.11 m) and 83 % of the teacher (2.27 m), with 0.50 aborted cycles per run (Table V), and exceeds SGT in one condition (2.59 vs. 2.35 m). The unregularized teacher is itself slightly faster than SGT, which we attribute to its rate-aware smoothing. The policy is not cheaper per tick in our implementation (0.73 vs. 0.53 ms): for 13 nodes the KKT solve is already inexpensive, so the value of the policy lies in replacing the model evaluation by a learned map, not in speed.

E. Suspension Integrity

With the filter the payload cables are slack 2.7 % of the time under SGT (35.1 % without) and 1.6 % under the policy (33.6 %), in all 10 conditions (Table VI); the mean payload-cable tension rises from 6.2 to 18.0 N (Fig. 1e). For SGT the taut suspension also makes tipping more effective: progress increases from 1.49 to 2.20 m (+48 %, better in 8 of 10 conditions) and aborted cycles fall from 1.1 to 0.0. For the policy the filter restores the suspension at unchanged progress (1.32 vs. 1.29 m). It does not reduce peak loads: in 3 of 20 filtered runs the payload came closer than 0.08 m to a rod (minimum 0.008 m), in 8 runs an outer cable exceeded 150 N, and the 99th-percentile payload acceleration of SGT increased—the inertial events excluded by Remark 2. The filter adds 3.6 ms per control step.

VII. CONCLUSION AND FUTURE PERSPECTIVES

We made tensegrity-rover simulation auditable with a non-minimal Lagrangian DAE, a minimum-norm constraint-and-energy corrector, passive contact with a potential and a work ledger that measures the unexplained energy of every step before it is corrected; it exposes the kilojoule-scale projection exchange and the non-physical aids of a conventional engine. On this basis, sensitivity-gradient tipping rolls a six-bar rover with a heavy payload using cables only, where reduced-order schedulers of a fixed primitive do not. The same sensitivity model serves as the teacher of MPC-guided policy search, whose per-cable policy reaches 89 % of SGT's progress, and as the prediction model of a suspension-integrity filter that keeps the payload cables taut and makes SGT 48 % faster.

TABLE V
MPC + GPS (MPC-GUIDED POLICY SEARCH): HELD-OUT EVALUATION, 4 CONDITIONS (LEVELS 1 AND 3, STRAIGHT AND ±6 M OFFSET GOALS, UNSEEN SEEDS), 20 S CABLE-ONLY RUNS, MEAN ± STD.

Method	Δy (m)↑	Tips↑	Aborts↓	t_{ctrl} (ms)↓
SGT	2.11 ± 0.34	11.0 ± 1.4	0.25 ± 0.50	0.53 ± 0.05
MPC teacher ($\lambda = 0$)	2.27 ± 0.32	10.8 ± 1.3	0.25 ± 0.50	0.63 ± 0.04
MPC + GPS policy	1.89 ± 0.47	11.5 ± 1.7	0.50 ± 0.58	0.73 ± 0.05

Limitations. All results are in simulation with contact and cable parameters not identified on hardware; the ledger bounds the energy contributed by integrator, projections and aids but does not validate the model against reality. SGT is quasi-static, re-plans at 20 Hz (about $0.1\,\mathrm{m/s}$), is sensitive to heavy-tailed measurement noise and zig-zags because its edge choice is greedy per cycle. The policy imitates a teacher on the same model, needs one KKT solve per cycle and is not faster than SGT; the filter guarantees nothing during impacts.

Remark 3. The modelled shell is a regular icosahedron ($d/h = 1/\varphi$). A form-finding analysis with the TsgFEM equilibrium matrix [21] shows that this shape admits no self-stress state, whereas the expanded octahedron ($d/h = 0.5$) admits one; the modelled rest geometry is therefore not a prestressed equilibrium, and part of the observed sag may originate there.

Future perspectives. A second-order barrier on the DAE dynamics for impact-safe payload clearance; repeating the studies on the form-found shape; a robust estimator in front of the planner and a heading-aware edge selection; and hardware validation.

REPRODUCIBILITY

`npm run experiments && npm run paper:data && npm run paper:ext` reruns all experiments and regenerates every table, figure and number (https://github.com/kapildev1012/3d).

APPENDIX
TERRAIN GENERALIZATION

Table VII reports SGT and the support-face heuristic on the rougher Levels 4 and 7 (3 seeds per cell). SGT keeps 0.074 m/s on Level 4 and 0.087 m/s on Level 7 with 100 % success; the ledger residual stays below 4×10^{-2} J.

TABLE VI

SUSPENSION-INTEGRITY FILTER, 10 PAIRED CONDITIONS (3 SEEDS ON LEVEL 1, 2 ON LEVEL 3, PUSH, IMPULSE, PAYLOAD $+30\,\%$, 100 MS DELAY, CABLE STIFFNESS $-20\,\%$), 20 S CABLE-ONLY RUNS, MEAN $\pm$ STD OVER CONDITIONS. "BETTER" COUNTS PAIRED CONDITIONS IN WHICH THE FILTER IMPROVED THE METRIC. METRICS EXCLUDE THE FIRST SECOND (INITIAL SETTLING).

	SGT			MPC + GPS policy		
Metric	off	on	better	off	on	better
Δy (m)$\uparrow$	1.49 ± 0.39	2.20 ± 0.64	8/10	1.32 ± 0.64	1.29 ± 0.54	4/10
Payload-cable slack time (%)$\downarrow$	35.1 ± 2.9	2.7 ± 0.6	10/10	33.6 ± 2.4	1.6 ± 0.6	10/10
Mean payload-cable tension (N)$\uparrow$	6.2 ± 0.5	18.0 ± 1.0	10/10	6.9 ± 0.6	17.5 ± 0.8	10/10
Aborted tip cycles$\downarrow$	1.1 ± 1.1	0.0 ± 0.0	6/10	1.5 ± 1.4	1.7 ± 1.4	3/10
Payload rattle RMS (mm)$\downarrow$	18.6 ± 1.0	15.0 ± 2.5	9/10	22.4 ± 1.0	27.5 ± 4.6	3/10
Payload accel. 99th pct (g)$\downarrow$	2.50 ± 1.08	3.18 ± 1.70	4/10	2.04 ± 1.12	1.71 ± 0.66	5/10
Min payload–rod clearance (m)$\uparrow$	0.102 ± 0.018	0.090 ± 0.035	5/10	0.105 ± 0.014	0.101 ± 0.033	4/10
Max outer-cable tension (N)$\downarrow$	137 ± 81	179 ± 79	3/10	122 ± 65	105 ± 52	7/10
t_{ctrl} (ms)$\downarrow$	0.64 ± 0.11	3.60 ± 0.21	–	0.79 ± 0.12	3.64 ± 0.23	–

TABLE VII

TERRAIN GENERALIZATION (3 SEEDS PER CELL).

Terrain / method	$\bar{v}$ (m/s)	RMSE (m)$\downarrow$	Succ. (%)	max $\lvert\varepsilon\rvert$ (J)
Level 1: SGT	0.080 ± 0.029	0.258 ± 0.040	100	1.0×10^{-9}
Level 1: Heuristic	0.004	0.083	0	9.9×10^{-10}
Level 4: SGT	0.074 ± 0.019	0.275 ± 0.044	100	1.0×10^{-9}
Level 4: Heuristic	0.004	0.083	0	9.9×10^{-10}
Level 7: SGT	0.087 ± 0.020	0.229 ± 0.008	100	1.5×10^{-2}
Level 7: Heuristic	0.004	0.083	0	9.9×10^{-10}

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